

Insulation Resistance (Megger) Test

This test determines the integrity of the stator winding insulation to ground. It is critical for identifying moisture ingress or thermal breakdown of the winding enamel.

Testing Parameters

- **Test Voltage:** Use 500V DC for motors rated up to 600V. Use 1000V DC for motors rated 2300V and above.
- **Duration:** Apply voltage for exactly 1 minute to obtain the IR value.
- **Minimum Acceptable Value:** A general rule of thumb is 1 Megohm per 1000V of operating voltage plus 1 Megohm. For a standard 480V motor, the minimum acceptable reading is 1.5 Megohms.

If readings fall below the minimum threshold, bake the stator in an industrial oven to remove moisture, and re-test before condemning the winding.

